KIW 53 (4/25/26)

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(Riemann) $\quad \zeta(s) = \sum_{n=1}^{\infty} |\mathbb{Z}/n\mathbb{Z}|^{-s}$

(Weil) fix a smooth curve mod p: say, $f(x,y) \equiv 0$

$$\zeta_f(s) = \sum_{\substack{I \trianglelefteq R_f \\ |R_f/I|<\infty}} |R_f/I|^{-s} = \sum_{\ell \geq 0} p^{-\ell s} \, |\{I \trianglelefteq R_f \mid \dim_{\mathbb{F}_p} R_f/I = \ell\}|$$

where $R_f = \mathbb{F}_p[x,y]/(f)$

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(1) $\quad \zeta_f(s) = \dfrac{\Phi_f(p^{-s})}{1-p^{1-s}}$ where $\Phi_f(q) \in \mathbb{Z}[q]$ of deg $2g_C$

(2) $\quad$ the roots of $\Phi_f(p^{-s})$ all lie on the line $\mathrm{Re}(s) = \frac{1}{2}$

<u>Ex</u> take $f(x,y) = y^2 - x^3 - c$ where $c \neq 0$

here, $\Phi_f(q) = 1 - aq + q^2$ where $|a| \leq 2p^{1/2}$

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if the curve is singular, then the R.H. (2) can fail

suppose f has a unique singularity at $(0,0)$

$$\text{here,} \qquad \zeta_f(s) = (\text{stuff})\, \hat{\zeta}_f(s)$$

(stuff) satisfies R.H.

$$\hat{\zeta}_f(s) = \sum_{\substack{I \subseteq \hat{R}_f \\ |\hat{R}_f/I|<\infty}} \frac{1}{|\hat{R}/I|^s} \text{ where } \hat{R}_f = \mathbb{F}_p[\![x,y]\!]/(f)$$

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in general, $\hat{\zeta}_f(s) = \dfrac{\hat{\Phi}_f(p^{-s})}{1 - p^{-s}}$ for some $\hat{\Phi}_f(q) \in \mathbb{Z}[q]$

<u>Ex</u> $f = y^2 - x^3$: $\hat{\Phi}_f(q) = 1 + pq^2$

<u>Ex</u> $f = y^3 - x^4$: $\hat{\Phi}_f(q) = 1 + pq^2 + p^2q^3 + p^2q^4 + p^3q^6$

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now take $f(x, y)$ over $\mathbb{C}$: the singularity defines a link L_f

$$\begin{array}{cccccc} & L_f := \{f = 0\} & \cap & S^3 & \subseteq & \mathbb{C}^2 \\ \dim & 1 = & 2 & + \ 3 & - & 4 \end{array}$$

<u>Example</u> if $f = y^n - x^m$ then $L_f = T_{n,m}$

over $\mathbb{F}_p$, seems that $\mathbb{P}_{L_f}(0, q, (pq)^{1/2}) = \hat{\Phi}_f(q)$

where $\mathbb{P}_L(a, q, t)$ is a normalization of $\dim_{a,q,t} \mathrm{HHH}(L)$

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now take $\hat{R}_f = \mathbb{C}[\![x, y]\!]/(f)$

<u>Conj</u> (Oblomkov–Rasmussen–Shende)

$\mathbb{P}_{L_f}(0, q, (tq)^{1/2}) = \sum_{\ell \geq 0} q^\ell \chi(t^{1/2}, \mathcal{H}_f^\ell)$ for

$$\text{Hilbert schemes} \quad \mathcal{H}_f^\ell = \{I \trianglelefteq \hat{R}_f \mid \dim_{\mathbb{C}} \hat{R}_f/I = \ell\}$$

$\mathbb{P}_{L_f}(a, q, (tq)^{1/2}) = \sum_{\ell, k \geq 0} q^\ell a^k t^{\binom{k}{2}} \chi(t^{1/2}, \mathcal{H}_f^{\ell,k})$ for

$$\mathcal{H}_f^{\ell,k} = \{(I, J) \in \mathcal{H}_f^\ell \times \mathcal{H}_f^{\ell+k} \mid xI + yI \subseteq J \subseteq I\}$$

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<u>Thm</u> (Maulik) true in HOMFLYPT limit $t \to -1$

proof by induction on blowups + "wall crossing"

<u>Observation</u> nicest case is $f = y^n - x^m$ for n, m coprime:

$\mathbb{P}_{T_{n,m}}$	$\mathcal{H}_f^{\ell,k}$
$n \times m$ Dyck paths	monomial ideals of $\hat{R}_f \simeq \mathbb{C}[\![t^n, t^m]\!]$

even here: unknown how to match combinatorics

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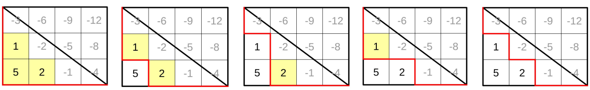

$$\mathbb{P}_{T_{n,m}}(a, q, t) = \sum_D q^{a(D)} t^{c(D)} \prod_{\square \nearrow D} (1 + a t^{e_{D,\square}})$$

<u>Ex</u> for $(n, m) = (3, 4)$,

$q^3 t^3 \cdot 1 + (q^2 t^2 + q t^2 + q t) \cdot (1 + at) + 1 \cdot (1 + at)(1 + at^2)$

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$\mathcal{H}_f^{\ell,m}$ stratified by monomial <u>ideals</u> of $\hat{R}_f \simeq \mathbb{C}[\![t^n, t^m]\!]$

whereas D defines <u>submodule</u> $M_D := \sum_{\square \to D} \mathbb{C}[\![t^n]\!] t^{\square} \subseteq \mathbb{C}[\![t]\!]$

(Kivinen–T) let $\hat{S}_f \supseteq \hat{R}_f$ be the <u>normalization</u>

$$\text{Quot schemes} \quad \mathcal{Q}_f^{\ell} = \{M \trianglelefteq \hat{S}_f \mid \dim_{\mathbb{C}} \hat{S}_f / M = \ell\}$$

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<u>Thm</u> (Kivinen–T) suppose $f = y^n - x^m$

if either n, m coprime or $n \mid m$:

$$(*) \qquad \mathbb{P}_{L_f}(a, q, t^{1/2}) = \sum_{\ell, m \geq 0} q^\ell a^m t^{\binom{m}{2}} \chi(t^{1/2}, \mathcal{Q}_f^{\ell, m})$$

for $\mathcal{Q}_f^{\ell,k} \subseteq \mathcal{Q}^\ell \times \mathcal{Q}^{\ell+k}$ analogous to $\mathcal{H}_f^{\ell,k}$

<u>Rem</u> implies motivic substitution relating $\mathcal{H}_f^{\ell,k}$ and $\mathcal{Q}_f^{\ell,k}$

the "Hilb-vs-Quot Conjecture"

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in general: components of L_f are <u>iterated torus knots</u>

next hardest: $f(x,y)$ with root $y = x^{m/n} + x^{(md+1)/(nd)}$

$\hat{R}_f = \mathbb{C}[\![t^{nd}, t^{md} + t^{md+1}]\!]$ $\quad L_f = (n, mnd+1)$ cable of $T_{n,m}$

<u>Ex</u> $f = y^4 - x^7 + x^6 - 4x^5 y - 2x^3 y^2$

$(d, n, m) = (2, 2, 3)$: first case not isotopic to torus knot

<u>Thm</u> (T in progress) $(*)$ holds in these cases

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(Caprau–González–Hogancamp–Mazin)

$\mathbb{P}_{L_f}$ via $nd \times md$ Dyck paths

(Gorsky–Mazin–Oblomkov)

technique to stratify $\mathcal{Q}_f^\ell$, $\mathcal{Q}_f^{\ell,k}$ combinatorially